COMPONENT SIDE — USB at left

70 × 50 mm • 24 × 18 holes • 2.54 mm pitch • all point-to-point wires are on rear.

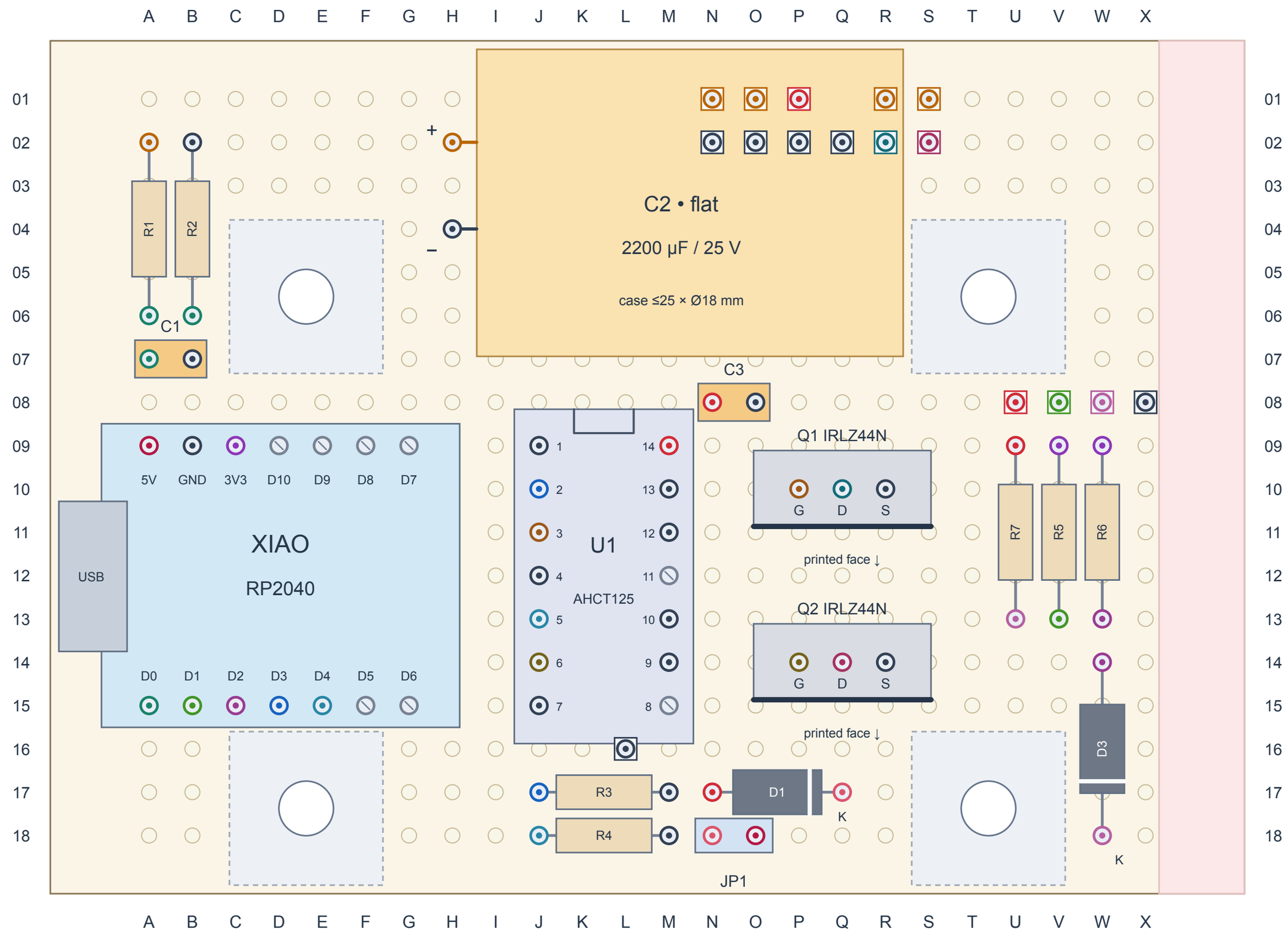
Solid dots: through-hole leads. Square ports: rear-only cable attachment pads.

Coordinates are assigned by this drawing; supplier lettering may be on the opposite face.

R1 100k · R2 22k · R3/R4 10k · R5/R6 4.7k · R7 3.3k

C1/C3 0.1 µF · D1/D3 NTE585 · JP1 removable shunt (remove for USB)

D2 stays at the pump; C4 stays at the Hall sensor. Neither mounts on this board.



U1 notch ↑ • Q1/Q2 printed faces ↓ • D1 stripe → Q17 • D3 stripe → W18 • C2 + H02 / − H04

Grey squares: 9 × 9 mm mounting keep-outs. Drill centres: x=15/55, y=15/45 mm from front top-left.

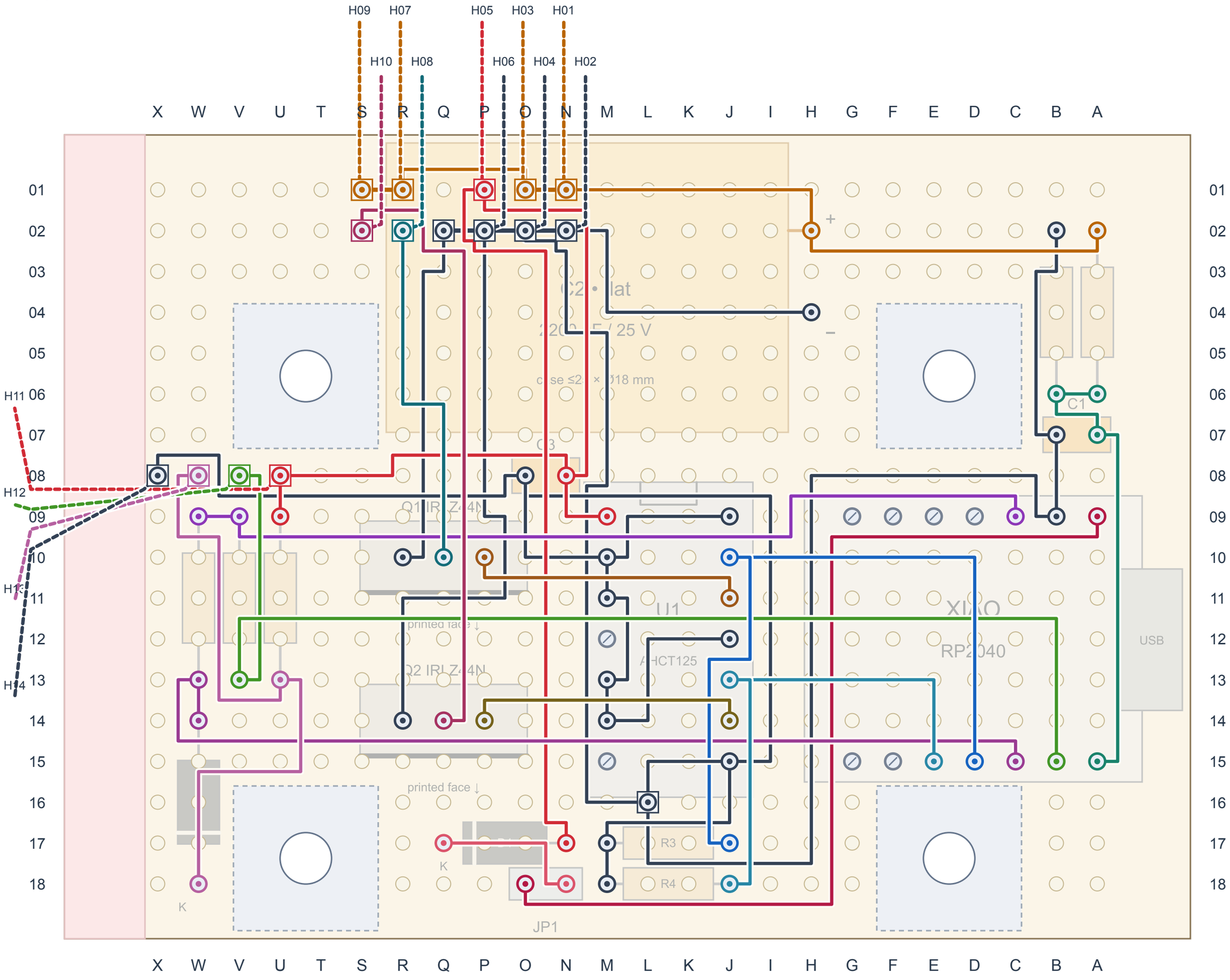
Grid offsets are nominal. Verify the board and dry-fit C2 before drilling or soldering. This is not a 1:1 drill template.

SOLDER SIDE — mirrored

Flip the board left ↔ right; top stays top. Pad names do not change.

Insulated wires cross without joining. Solder only at marked endpoints; bends are not joints.

Only W01, W02 and W05–W07 are short bare bus links. All other links are insulated.



Solder-side drawing shows every wire; use the wire schedule or interactive net filter to trace crowded areas.

Grey squares: 9 × 9 mm mounting keep-outs. Drill centres: x=15/55, y=15/45 mm from front top-left.

Grid offsets are nominal. Verify the board and dry-fit C2 before drilling or soldering. This is not a 1:1 drill template.

CABLE CONNECTIONS AND COLOUR KEY

Dashed leads on the solder-side drawing. Attach to the rear copper pads.

H01	N01	S1 switched output
H02	N02	Battery negative / J1 sleeve
H03	O01	Buck IN+
H04	O02	Buck IN−
H05	P01	Buck OUT+ (set to 5.0 V)
H06	P02	Buck OUT−
H07	R01	Pump + / D2 striped end
H08	R02	Pump − / D2 unstriped end
H09	S01	LED + (JST SM)
H10	S02	LED − (JST SM)
H11	U08	Lock +5 V: A3144 pin 1 / C4
H12	V08	Lock HALL: A3144 pin 3
H13	W08	Lock SW: microswitch NO
H14	X08	Lock GND: Hall pin 2 / COM / C4

OFF-BOARD CONNECTIONS

Battery + → J1 centre and S1 input. S1 output → H01. Battery − → J1 sleeve and H02.
Leave J1 switched contact unused. Set the buck output to 5.0 V before connecting H05/H06.
D2 NTE586 across pump terminals: stripe to pump +. C4 0.1 µF across A3144 pins 1–2.
A3144: printed face toward you, leads down; pins 1/2/3 = +5V/GND/HALL.
Twist switch COM → GND, NO → SW; NC unused. Lock cable uses the JST PH pair.
JP1: fit shunt for battery operation; remove for USB and follow build-guide power procedure.

- Switched battery
- Ground
- Regulated 5 V
- D1 → JP1
- JP1 → XIAO 5V
- 3.3 V
- Battery sense
- Hall signal
- Twist GPIO
- Switch contact
- Pump PWM
- LED PWM
- Pump gate
- LED gate
- Pump return
- LED return

Cable H02 is an identifier, not a pad: it attaches to N02. Pad H02 is C2 positive.

WIRE-BY-WIRE SOLDER SCHEDULE

Use component-side pad names on both faces. All links on rear; solder only at their two endpoints.

W01	N01 → O01	VLOAD	BARE link	W27	N08 → U08	5V_REG	insulated
W02	R01 → S01	VLOAD	BARE link	W28	U08 → U09	5V_REG	insulated
W03	O01 → R01	VLOAD	insulated	W29	P01 → N17	5V_REG	insulated
W04	H02 → N01	VLOAD	insulated	W30	Q17 → N18	AFTER_D1	insulated
W05	N02 → O02	GND	BARE link	W31	O18 → A09	XIAO_5V	insulated
W06	O02 → P02	GND	BARE link	W32	C09 → V09	3V3	insulated
W07	P02 → Q02	GND	BARE link	W33	V09 → W09	3V3	insulated
W08	H04 → N02	GND	insulated	W34	A06 → B06	ADC	insulated
W09	R10 → Q02	GND	insulated	W35	B06 → A07	ADC	insulated
W10	R14 → P02	GND	insulated	W36	A07 → A15	ADC	insulated
W11	L16 → O02	GND	insulated	W37	A02 → H02	VLOAD	insulated
W12	B09 → L16	GND	insulated	W38	V13 → V08	HALL	insulated
W13	B09 → B07	GND	insulated	W39	V13 → B15	HALL	insulated
W14	B07 → B02	GND	insulated	W40	W13 → C15	TWIST	insulated
W15	J15 → M17	GND	insulated	W41	W13 → W14	TWIST	insulated
W16	M17 → M18	GND	insulated	W42	U13 → W18	SW	insulated
W17	J15 → O08	GND	insulated	W43	U13 → W08	SW	insulated
W18	M14 → M13	GND	insulated	W44	D15 → J10	PWM_P	insulated
W19	M13 → M11	GND	insulated	W45	J10 → J17	PWM_P	insulated
W20	M11 → M10	GND	insulated	W46	E15 → J13	PWM_L	insulated
W21	J09 → M10	GND	insulated	W47	J13 → J18	PWM_L	insulated
W22	J12 → M14	GND	insulated	W48	J11 → P10	GATE_P	insulated
W23	O08 → M10	GND	insulated	W49	J14 → P14	GATE_L	insulated
W24	X08 → O08	GND	insulated	W50	Q10 → R02	DRAIN_P	insulated
W25	P01 → N08	5V_REG	insulated	W51	Q14 → S02	DRAIN_L	insulated
W26	N08 → M09	5V_REG	insulated	W52	L16 → J15	GND	insulated

Q02 and L16 are wire junction pads; no component goes in these holes.

Crossings and bends are not joints. Keep all other bare metal inside a solder fillet.